Magnetic fields

antocciphysics | TEACHER KEY | CPA

Learning target: Interpret magnetic field patterns and compass direction.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 9; 3 course days later.*
5 A in each branch.

02. Explain a worked example (worked)
Arrows leave N and enter S outside; compass arrows tangent to field.

03. Look and explain (model)
External arrows N to S; compass north end tangent to the local field.

04. Build the solution (practice)
N-N repels; N-S attracts.

05. Use the idea (reason)
A nonzero field at one point has one net direction; crossing lines assign two different directions.

06. Choose a method (mixed) *Review: Unit 6, Day 18; 12 course days later.*
Bends toward the normal; frequency stays the same at a stationary boundary.

07. Exit check (exit)
A stronger field in a consistently drawn field map.

08. STRETCH: Apply it another way (stretch)
Each ordinary piece has both N and S; cutting does not isolate a single pole.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 9; 3 course days later.*
Parallel 3 Ω; total 8 Ω; supply current 2 A.

02. Explain a worked example (worked)
Arrows leave N and enter S outside; compass arrows tangent to field.

03. Represent and interpret (model)
External arrows N to S; compass north end tangent to the local field.

04. Solve and justify (practice)
Opposing fields may nearly cancel; Earth’s field, friction and measurement sensitivity matter, so it does not prove all fields vanish.

05. Analyze the reasoning (reason)
A nonzero field at one point has one net direction; crossing lines assign two different directions.

06. Choose a method (mixed) *Review: Unit 6, Day 18; 12 course days later.*
Thermal peak becomes 1/4 by Wien’s law; sound wavelength becomes 1/6 by $\lambda=v/f$. Different models and assumptions.

07. Exit check (exit)
No; each piece has north and south poles.

08. STRETCH: Apply it another way (stretch)
Like poles repel with field bending away from the central region; N facing S attracts with connecting field lines. Qualitative superposition, no lines crossing.